

BloodConnect Analytics Accuracy and Validation Report

A detailed, plain-language explanation of the predictive and prescriptive analytics tests, calculations, formulas, algorithms, and results.

System	BloodConnect
Dataset	Controlled Synthetic Historical Blood Inventory and Utilization Dataset
Coverage	January 1 to December 31, 2025
Test rerun	September 13, 2026
Evaluation unit	Hospital + blood group + blood component

Important: This report validates the software against controlled synthetic data. It does not prove accuracy in real hospitals and should not be presented as clinical validation.

REPORT GUIDE

What this report answers

BloodConnect performs three different jobs. Each job needs its own measure; combining them into one percentage would hide important weaknesses.

Predictive: demand

Estimates how many blood units will be used during the next seven days.

Main measures: MAE, RMSE, WAPE and MAPE. Lower error is better.

Predictive: shortage

Decides whether an actual shortage will occur during the next seven days.

Main measures: accuracy, precision, recall, F1 and balanced accuracy.

Prescriptive: recommendations

Suggests how to cover requests using local stock, central stock, another hospital, or donor/external coordination.

Main measures: plan feasibility and percentage of requested units covered. This is not a prediction accuracy score.

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How to read percentages: A high accuracy value can still hide missed shortages when most observations are "no shortage." For that reason, recall and balanced accuracy are shown beside accuracy.

1. EXECUTIVE SUMMARY

Overall test results

83.64%

Demand score calculated as 100 - WAPE

78.19%

Shortage classification accuracy

100%

Prescriptive plans technically feasible

Area tested	Result	Plain-language meaning
Data integrity	18 / 18 passed	The controlled files were internally consistent for all implemented checks.
Demand MAE	0.93 units	A forecast was wrong by about 0.93 unit on average.
Demand WAPE	16.36%	Total absolute error was 16.36% of total actual fulfilled usage.
Shortage precision	50.87%	About half of shortage alerts corresponded to an actual shortage.
Shortage recall	36.99%	The engine detected about 37 of every 100 actual shortages.
Coverage after recommendations	88.68%	The simulated plan covered 36,198 of 40,817 requested units.
Scenario condition detection	71.43%	10 of 14 supported controlled conditions were correctly identified.

Plain-language verdict: Demand forecasting performs better than the naive comparison. Shortage alerts have substantially fewer false alarms than the old rule, but the engine still misses many shortages. The transfer plan obeys tested stock and reserve constraints, although it improves coverage by less than one percentage point.

No single overall accuracy: 83.64%, 78.19%, and 100% answer different questions. They should be reported separately in the capstone.

2. DATASET AND QUALITY CHECKS

What data was tested?

The supplied dataset is a controlled synthetic year of blood-bank activity. It contains structured patterns such as normal demand, rising demand, spikes, shortages, overstock, near-expiry stock, transfer opportunities and reduced supply.

6

Hospitals

8

Blood groups

3

Components

File	Rows	Main use in testing
inventory_history.csv	52,560	Daily opening, receipts, fulfillment, expiry and closing stock.
blood_usage_history.csv	52,560	Daily requested, fulfilled and transfused quantities.
donations_supply.csv	52,560	Incoming supply reconciliation.
hospital_requests.csv	33,017	Priorities, requested units and fulfilled units.
blood_units.csv	49,190	Unit-level receipt, use and expiration chronology.
expected_test_scenarios.csv	15	Known scenario labels and expected forecast values.

Independent checks performed

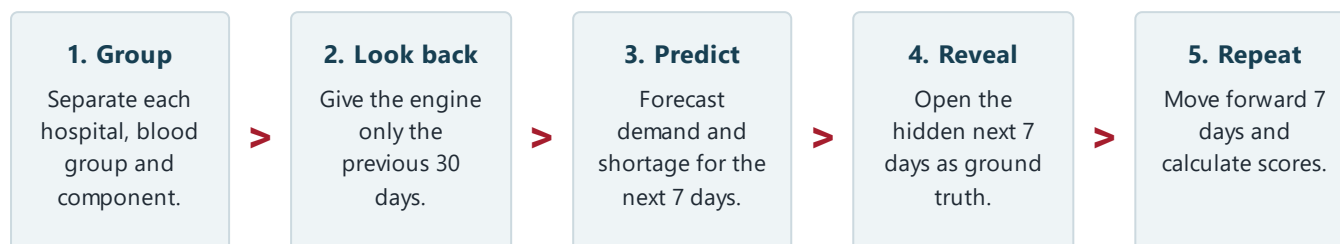
- Expected row counts and unique identifiers
- No negative inventory values
- Daily inventory arithmetic
- Day-to-day stock continuity
- Fulfilled units do not exceed requested units
- Transfused units do not exceed fulfilled units
- Requests reconcile with usage totals
- Supply reconciles with inventory receipts
- Unit use and expiration reconcile with inventory
- Units are not used on or after expiry

Result: 18 of 18 implemented integrity checks passed. The test script recalculated these checks and did not simply trust the supplied validation text.

Synthetic generation used fixed seed 20250909. This improves reproducibility but does not make the data real-world evidence.

3. TESTING PROCESS

How the accuracy test was performed



Why there were 6,768 forecast cases

6 hospitals x 8 blood groups x 3 components = 144 series
 144 series x 47 weekly forecast origins = 6,768 forecast cases

Rolling-origin validation

Every prediction was made at a historical date. The following seven days were withheld until after the prediction. The origin then moved forward by seven days. This is closer to real forecasting than randomly mixing past and future rows.

Input available to the engine	Hidden outcome used only for scoring
Previous fulfilled and requested records	Next seven days of fulfilled usage
Inventory available immediately before forecast date	Whether any requested units were unfulfilled
Batch expiration dates and current stock	Actual condition over the hidden horizon

Protection against leakage: Future rows were not supplied to the predictive engine. For the separate prescriptive stress test, hidden seven-day demand was converted to pending workload only after predictive scoring.

Ground-truth definitions

- **Demand actual:** the total units fulfilled during the hidden seven days.
- **Actual shortage:** requested units were greater than fulfilled units during that hidden period.
- **Engine alert:** only critical, out-of-stock, or unusable-near-expiry states counted as a positive shortage alert.

4. PREDICTIVE ANALYTICS - DEMAND

Algorithm and logic

The demand module uses a moving historical average. It totals fulfilled units from the previous 30 days, converts that total into a daily average, and multiplies by seven.

Main formula

```
Average daily usage = fulfilled units in previous 30 days / 30  
Forecast demand for next 7 days = average daily usage x 7
```

Trend and unusual-change rules

- Compare the current 30-day total with the preceding 30-day total.
- Increase greater than 10%: increasing trend.
- Decrease greater than 10%: decreasing trend.
- Otherwise: stable trend.
- Current usage at least 1.6 times the previous period: spike.
- Current usage at most 0.5 times the previous period: drop.

Representative application snippet - simplified excerpt

```
const averageDailyUsage = currentUnits / periodDays  
const expectedDemandNext7Days = averageDailyUsage * 7  
  
const percentChange = previousUnits > 0  
  ? ((currentUnits - previousUnits) / previousUnits) * 100  
  : currentUnits > 0 ? 100 : 0  
  
const trend = percentChange > 10 ? 'increasing'  
  : percentChange < -10 ? 'decreasing'  
  : 'stable'
```

Important limitation: The numerical demand forecast uses fulfilled units. When stock is already scarce, fulfilled usage can be lower than true requested demand. The shortage module partly compensates by separately examining requested and unmet units.

Comparator

The engine was compared with a simple naive method: "next week will equal last week." BloodConnect achieved lower MAE and WAPE than this comparator.

5. DEMAND ERROR FORMULAS AND RESULTS

How forecast error was calculated

Measure	Formula	Result
MAE	$\text{sum}(\text{forecast} - \text{actual}) / \text{number of forecasts}$	0.931 units
RMSE	$\text{square root of mean}((\text{forecast} - \text{actual})^2)$	1.328 units
WAPE	$\text{sum}(\text{forecast} - \text{actual}) / \text{sum}(\text{actual}) \times 100$	16.363%
MAPE	$\text{mean}(\text{forecast} - \text{actual} / \text{actual}) \times 100$	26.721%
Bias	$\text{mean}(\text{forecast} - \text{actual})$	-0.025 units
Within 20%	cases with error no greater than 20% / all cases	51.995%

Accuracy-like conversions

WAPE-derived score = $100\% - 16.36295\% = 83.63705\%$

MAPE-derived score = $100\% - 26.72118\% = 73.27882\%$

Why the two numbers differ: WAPE combines all absolute errors before dividing by total actual usage, so higher-volume cases have more influence. MAPE calculates a percentage for every nonzero case and then averages those percentages, so low-volume cases can have a much larger effect.

Zero actual values

MAPE cannot divide by zero. Therefore, 1,148 cases with zero actual usage were excluded from MAPE, leaving 5,620 included cases. MAE, RMSE and WAPE used all 6,768 cases.

Comparison with naive last-week forecast

BloodConnect WAPE		16.36%
Naive WAPE		18.78%

Relative WAPE improvement = $(18.78099 - 16.36295) / 18.78099 \times 100 = 12.88\%$

The bar lengths are scaled to a 20% error reference. Shorter is better.

6. PREDICTIVE ANALYTICS - SHORTAGE

Algorithm and decision logic

The shortage module combines requested demand, actual fulfilled usage, current stock, expiration dates, recent unmet demand and repeated under-fulfillment.

Core calculations

Expected 7-day demand = (requested units in previous 30 days / 30) x 7

Coverage ratio = usable stock / expected 7-day demand

Recent unmet ratio = unmet requested units / total requested units

How near-expiry stock is handled

Stock expiring within seven days is not automatically discarded. The engine counts only the quantity that could reasonably be consumed before each batch expires. This avoids both overestimating and unnecessarily ignoring near-expiry inventory.

Alert policy

Condition	System state	Confirmed alert?
No usable stock and historical demand exists	Critical / out of stock	Yes
All stock expires before expected use	Critical / near-expiry only	Yes
Coverage ratio is 0.20 or lower	Critical	Yes
At least 2 underfilled events and unmet ratio at least 0.65	Persistent critical underfill	Yes
Coverage below 1.00 or unmet ratio at least 0.55	Low - monitor	No

Representative application snippet - simplified excerpt

```
const averageDailyDemand = historicalDemand / usageWindowDays
const expectedDemandNext7Days = averageDailyDemand * 7
const coverageRatio = usableStock / expectedDemandNext7Days

if (usableStock === 0) shortageAlert = true
else if (coverageRatio <= 0.20 || persistentUnderfill)
  shortageAlert = true
else if (coverageRatio < 1 || underfillWarning)
  status = 'low - monitor'
```


7. SHORTAGE ACCURACY AND CONFUSION MATRIX

What the 78.19% score contains

	Actual shortage	No actual shortage
Engine alert	553 true positives	534 false positives
No engine alert	942 false negatives	4,739 true negatives

Accuracy

$$\begin{aligned}
 & (TP + TN) / \text{all cases} \times 100 \\
 & (553 + 4,739) / 6,768 \times 100 \\
 & = 78.19\%
 \end{aligned}$$

Specificity

$$\begin{aligned}
 & TN / (TN + FP) \times 100 \\
 & 4,739 / (4,739 + 534) \times 100 \\
 & = 89.87\%
 \end{aligned}$$

Precision

$$\begin{aligned}
 & TP / (TP + FP) \times 100 \\
 & 553 / (553 + 534) \times 100 \\
 & = 50.87\%
 \end{aligned}$$

F1 score

$$\begin{aligned}
 & 2TP / (2TP + FP + FN) \times 100 \\
 & 1,106 / 2,582 \times 100 \\
 & = 42.84\%
 \end{aligned}$$

Recall

$$\begin{aligned}
 & TP / (TP + FN) \times 100 \\
 & 553 / (553 + 942) \times 100 \\
 & = 36.99\%
 \end{aligned}$$

Balanced accuracy

$$\begin{aligned}
 & (\text{recall} + \text{specificity}) / 2 \\
 & (36.99\% + 89.87\%) / 2 \\
 & = 63.43\%
 \end{aligned}$$

Interpretation: Accuracy is only 0.28 percentage point above the 77.91% majority baseline. The engine is good at recognizing non-shortage cases, but it detected only 36.99% of actual shortages.

8. PRESCRIPTIVE ANALYTICS - RECOMMENDATIONS

Algorithm and decision sequence

The prescriptive module is a priority-based greedy allocation heuristic. It makes one safe choice at a time instead of searching every possible transfer combination.

1. Remove expired or invalid inventory from available stock.
2. Keep only pending, valid requests.
3. Sort requests by priority: critical, urgent, then normal. Older requests win ties.
4. Use stock already present at the destination first.
5. Calculate the remaining units needed.
6. Find sources with the exact same blood group and component.
7. Protect a 20-unit reserve at hospital sources; central inventory has no configured reserve in this rule.
8. Select the source with the largest sendable quantity.
9. Recommend no more than the remaining need or sendable stock.
10. Deduct the recommendation immediately so later requests cannot reuse the same units.

Prescriptive formulas

```
Units needed = max(0, units requested - destination stock used)
Hospital sendable stock = max(0, source stock - 20-unit reserve)
Suggested transfer = min(units needed, sendable stock)
```

Representative application snippet - simplified excerpt

```
activeRequests.sort(byPriorityThenOldest)

destinationUsed = Math.min(unitsRequested, destinationAvailable)
unitsNeeded = Math.max(0, unitsRequested - destinationUsed)

sendable = source === 'central'
  ? available
  : Math.max(0, available - reserveAtSourceUnits)

suggestedUnits = Math.min(sendable, unitsNeeded)
remainingStock[source] -= suggestedUnits
```

Recommendation choices: already covered locally; dispatch from central inventory; transfer from another hospital; or contact donors / coordinate external supply.

9. PRESCRIPTIVE TEST AND CALCULATIONS

What was tested?

For each of 47 weekly snapshots, the hidden seven-day requested total was converted into pending workload after the prediction had already been scored. The resulting plan was checked for invalid quantities, incompatible products, self-transfers, unavailable stock, expired stock, overcommitment and priority order.

47 / 47

Feasible plan runs

88.68%

Demand covered after recommendations

347

Additional units covered by transfers

Worked calculations

$\text{Feasibility} = 47 \text{ feasible runs} / 47 \text{ total runs} \times 100 = 100.00\%$

$\text{No-transfer coverage} = 35,851 / 40,817 \times 100 = 87.83\%$

$\text{Coverage after recommendations} = 36,198 / 40,817 \times 100 = 88.68\%$

$\text{Additional units covered} = 36,198 - 35,851 = 347 \text{ units}$

$\text{Absolute coverage increase} = 88.68\% - 87.83\% = 0.85 \text{ percentage point}$

What 100% means - and does not mean

It means: every generated plan obeyed the software constraints checked in the simulation.

It does not mean: every request was filled, the plan was mathematically optimal, or the recommendation was clinically validated.

Other controlled scenario results

Check	Result	Explanation
Expected forecast reproduction	14 / 15 (93.33%)	Numeric expected values matched within 0.011 unit.
Supported condition detection	10 / 14 (71.43%)	Includes shortage, overstock, expiry and transfer-source conditions.
Unsupported condition	1 excluded	Reduced supply is not an input to the current predictive engine.

10. CONCLUSIONS AND NEXT STEPS

Final assessment

Module	Assessment	Recommended capstone wording
Demand forecast	Promising in controlled test	WAPE was 16.36%, equivalent to an 83.64% WAPE-derived score, with MAE of 0.93 unit.
Shortage forecast	Needs improvement	Classification accuracy was 78.19%, but recall was only 36.99%; accuracy alone overstates performance.
Recommendations	Constraint-feasible	All 47 simulated plans were feasible and coverage reached 88.68%; clinical optimality was not tested.

Priority improvements

1. **Add incoming-supply forecasting.** Use recent donation receipts and scheduled donations to estimate supply during the seven-day horizon.
2. **Forecast true demand.** Train on requested demand as well as fulfilled usage so shortages do not hide unmet need.
3. **Calibrate alerts chronologically.** Use separate training, validation and final test periods. Choose alert thresholds on validation data only.
4. **Set multi-metric acceptance criteria.** Require at least 80% accuracy while also improving recall, F1 and balanced accuracy.
5. **Improve prescription realism.** Include travel time, transport capacity, crossmatching, component substitution policy, future arrivals and hospital-specific reserve targets.
6. **Validate with real de-identified operations.** Repeat the same locked protocol before making clinical or operational claims.

Study limitations

- The dataset is synthetic and contains controlled, deliberately constructed patterns.
- Shortage prevalence was 22.09%, so ordinary accuracy is strongly influenced by non-shortage cases.
- The current shortage model does not use future donation schedules or forecast incoming supply.
- Aggregate near-expiry counts were translated into test batches because the application expects batch expiration dates.
- The greedy transfer method is feasible but not proven optimal.

Recommended conclusion: BloodConnect passed functional validation on the controlled dataset and showed useful demand forecasting and feasible allocation behavior. Shortage detection requires further improvement, especially recall, before real-world deployment claims are appropriate.